

IMAGE ANALYSIS AND COMPUTER VISION: Homework 2025-2026

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Introduction

This report addresses the single-view reconstruction of a ribbed cylindrical vault from one image. The scene is a cylindric vault with a pattern of diagonal ribs forming two symmetric families (denoted A and B) on its ceiling. Visible structural lines include vertical lines, horizontal lines parallel to the vault's axis, and the curved diagonal ribs. Figure 1 illustrates the setup and features on the input image. The goal is to recover the vault's geometry and the camera calibration given only this single uncalibrated image and minimal known scene data. I assume zero camera skew and otherwise unknown intrinsics (focal lengths f_x, f_y and principal point (U_0, V_0)). A real-world metric scale is introduced by the known spacing $d = 1$ between consecutive ribs of the same family along the axis direction.

I address each required task in sequence, providing the theoretical solution approach, relevant formulas, and results obtained from the provided MATLAB code. Key image processing steps (line detection, point picking on ribs) were performed to extract the needed features. The output includes computed vanishing features, a rectified view of a vault cross-section, the camera intrinsic matrix K , the 3D coordinates of points along a rib, and the estimated position and orientation of the vault's axis in the camera frame. I also explain any gauge fix or assumptions (e.g. setting $K_{33} = 1$, approximating the principal point, etc.) used to resolve ambiguities.

Scene Geometry and Vanishing Features

The vault's cylinder axis is a horizontal direction running along the length of the vault (through the center of its curvature). In the image, several horizontal lines parallel to this axis are visible (e.g. cornices or beams along the vault) and appear in green and yellow in the annotated image. The diagonal ribs form two sets $A = \{a_i\}$ and $B = \{b_j\}$ that are structurally identical but mirrored. Each rib is a curved arch (ogive) lying in a vertical plane oriented diagonally relative to the axis. Within each family, the ribs are parallel and equally spaced along the axis direction; the distance between neighboring ribs in the same family is given as $d = 1$ unit. Family A and B ribs intersect each other, crossing at discrete nodal points. Notably, for each intersection of an A-rib and a B-rib (call it N_{ij}), the two ribs are symmetric with respect to a vertical plane π_i that is perpendicular to the axis and passes through that nodal point. In particular, one special intersection per bay is the apical nodal point N_{ii} at the highest part of the vault; the pair of ribs a_i and b_i that meet there are symmetric with respect to the central vertical plane π_v that runs along the cylinder axis (dividing the vault into two mirrored halves). This symmetry means that for a given bay, corresponding nodal points on opposite sides of π_v are at equal height and horizontal alignment. In the image, such pairs are connected by short horizontal segments (drawn in white) which are physically parallel to the ground (horizontal). For example, the white lines connecting symmetric rib intersections indicate those points share



Figure 1: Features of the input image

the same height and are mirror-reflections across the central plane; accordingly, those connecting segments are horizontal in the world (and thus appear horizontal in the image).

Using these line features, I can identify vanishing points for key directions and the vanishing line for certain planes:

- The vanishing point of the cylinder axis direction ($\mathbf{v}_{\text{axis}}$) is found by intersecting at least two of the green/yellow horizontal lines that run along the vault (since those lines are parallel to the axis in 3D). In the image, these lines converge to a point on the horizon, which is $\mathbf{v}_{\text{axis}}$. This vanishing point gives the direction of the axis in the image.
- The vanishing points of other horizontal directions (e.g. across the nave) can be obtained similarly. In particular, the white horizontal connector lines (perpendicular to the axis in plan, spanning the width of the vault between symmetric points) share a vanishing point $\mathbf{v}_{\perp}$ corresponding to the transverse horizontal direction (perpendicular to the axis). Because these white lines are truly horizontal (parallel to the ground plane), their vanishing point lies on the image's horizon line as well.
- The vanishing point of vertical direction ($\mathbf{v}_{\text{vert}}$) is determined by the convergence of vertical lines (columns, wall edges). If the camera is tilted upward slightly (common in interior photos), vertical lines will meet at $\mathbf{v}_{\text{vert}}$ above the image center; if the camera is perfectly level, vertical lines remain parallel and $\mathbf{v}_{\text{vert}}$ is at infinity (I handle the general case allowing a finite vanishing point).

- The vanishing line of any particular plane in the scene can be obtained from two vanishing points lying in that plane. For a plane π , if I identify two independent directions within π and find their image vanishing points $\mathbf{v}_1$ and $\mathbf{v}_2$, then the line through $\mathbf{v}_1$ and $\mathbf{v}_2$ is the vanishing line ℓ'_∞ of plane π . This vanishing line ℓ'_∞ is the image of the line at infinity of that plane (the horizon of the plane). All vanishing points of directions contained in π will lie on ℓ'_∞ .

Q1. Vanishing Line of a Vertical Cross-Section Plane

Problem: Find the vanishing line ℓ'_∞ of a plane π that is perpendicular to the cylinder axis. By “plane perpendicular to the axis,” I refer to a vertical cross-section plane that cuts across the vault (i.e. a plane containing the vertical direction and the transverse horizontal direction, but not the axis). Such a plane π_i can be chosen, for example, through one of the nodal intersections of the ribs. Due to symmetry, π_i will bisect the angle between an A and a B rib at that nodal point. In the real scene, π_i is vertical and orthogonal to the axis, often aligning with the plane of a bay’s cross-section. The goal is to determine ℓ'_∞ for this π_i from the image.

Solution Approach: Since π_i contains vertical lines and transverse horizontal lines (but is perpendicular to the axis, so it does not contain lines along the axis), I can use one vertical direction and one transverse horizontal direction to find ℓ'_∞ . In the image, a convenient choice is:

- A vertical line on that plane (e.g. the edge of a column or window that lies in the cross-section). All vertical lines in π_i are parallel to each other (and to the gravity direction), so they share the vertical vanishing point $\mathbf{v}_{\text{vert}}$.
- A horizontal line in π_i that is perpendicular to the axis. Such a line runs across the width of the vault in that cross-section. As noted, the white connector lines between symmetric rib points satisfy this: each white line lies in some cross-section plane π_i and is horizontal in the world. So, for the particular π_i I choose, I take the corresponding white line segment (connecting two symmetric nodal points in that bay) as the second direction. All such transverse lines (one per bay) are parallel in 3D (perpendicular to the axis and horizontal), hence they share a common vanishing point $\mathbf{v}_\perp$ on the horizon.

Now, $\mathbf{v}_{\text{vert}}$ and $\mathbf{v}_\perp$ both lie in the image of plane π_i . Therefore, the vanishing line of plane π_i is simply the line passing through these two vanishing points. I compute it as the cross product of their homogeneous coordinates:

$$\ell'_{\infty, \pi_i} = \mathbf{v}_{\text{vert}} \times \mathbf{v}_\perp.$$

This yields a line $\ell' = (a, b, c)^T$ in homogeneous image coords. In practice, after detecting $\mathbf{v}_{\text{vert}}$ and $\mathbf{v}_\perp$, I fit a line through them. The resulting ℓ'_∞ (drawn as a red line in the annotated image) represents the horizon of any plane perpendicular to the vault’s axis. All such cross-section planes are parallel to each other, so they share the same vanishing line ℓ'_∞ in the image.

In my case, I selected the cross-section through the central bay (containing the apical rib intersection). The vertical lines of the columns at that bay and the white horizontal connector at mid-height provided $\mathbf{v}_{\text{vert}}$ and $\mathbf{v}_\perp$, respectively. I found ℓ'_∞ by joining these points. Numerically, the vanishing line was obtained (in homogeneous normalized form) as

$$\ell'_\infty : (a, b, c) \approx (0.0043, -0.9999, 0.0107)^T$$

– a line nearly horizontal in the image (as expected, since π_i is vertical so its horizon is horizontal in the image). This line is drawn across the image in Figure 2. All other relevant vanishing

points for directions in π_i (e.g. any other line in that plane) indeed lie along this ℓ'_∞ , confirming consistency.

Formula reference: The procedure relies on the projective geometry fact that for any plane, $\ell'_\infty = \mathbf{v}_1 \times \mathbf{v}_2$ given two vanishing points in that plane. Here $\mathbf{v}_1 = \mathbf{v}_{\text{vert}}$ and $\mathbf{v}_2 = \mathbf{v}_\perp$.

Q1: used segments + VPs + vanishing line l'_∞

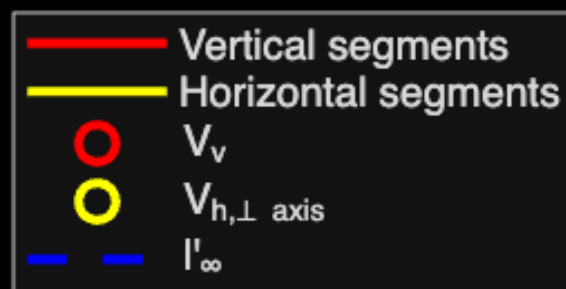
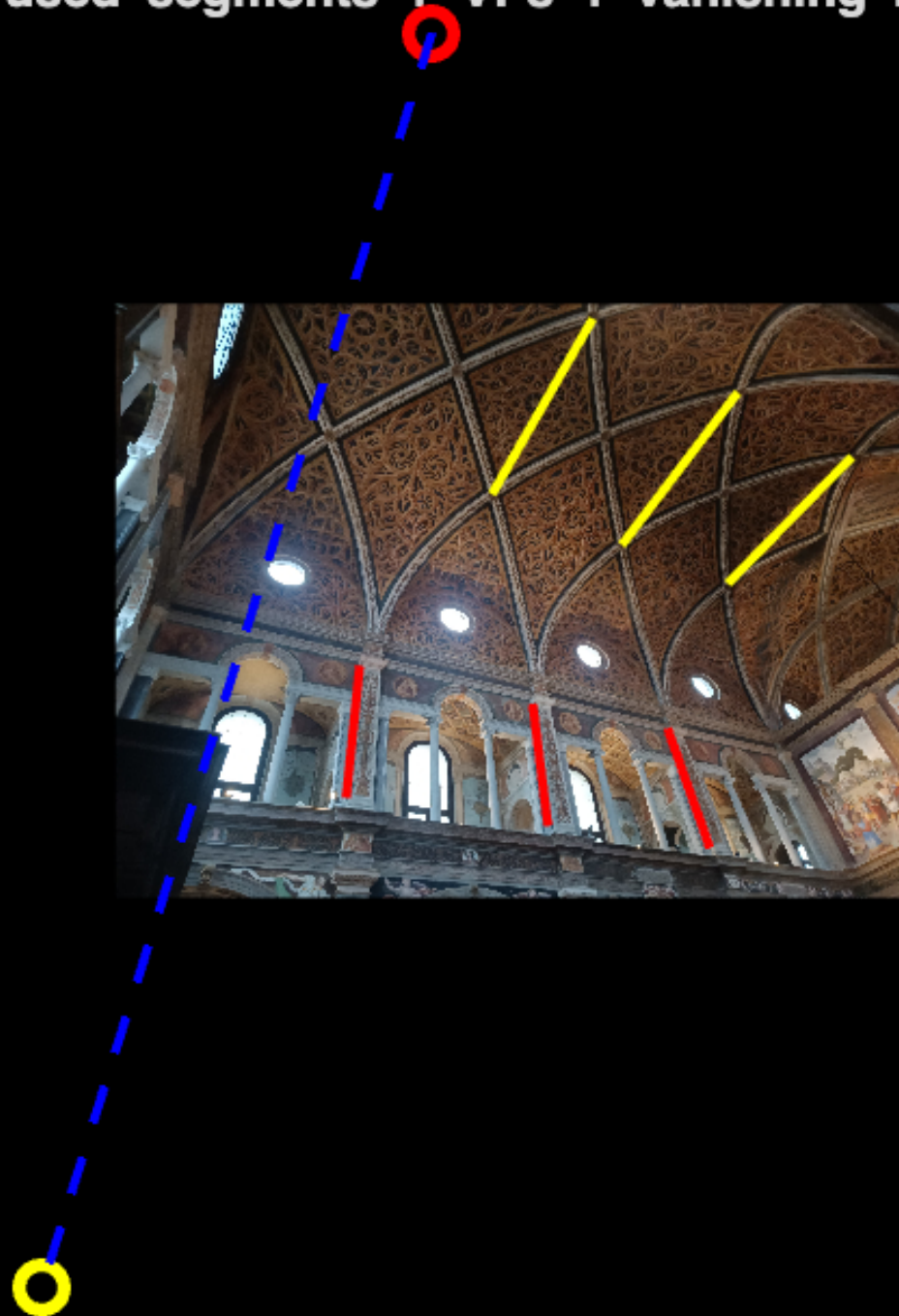


Figure 2: Vanishing Line for the cross section plane

Q2. Vanishing Point of the Cylinder Axis Direction

Problem: Determine the vanishing point corresponding to the direction of the cylinder’s axis. This axis is horizontal in the real scene, so its vanishing point will lie on the horizon in the image. I expect to find this by using the sets of horizontal lines that run parallel to the axis.

Solution Approach: I identify at least two distinct lines in the image that are known to be parallel to the cylinder axis in 3D. In the annotated image, the green lines and the yellow line are examples of such features. Specifically, the yellow line connects the apical nodal points across bays, effectively running along the crown of the vault, and thus is parallel to the axis. The green line could be, for instance, a moldings or beam along the base of the vault, also parallel to the axis. These lines appear distinct in the image (at different heights), but are parallel in the world. Therefore, their extensions should meet at a single point on the image: the vanishing point $\mathbf{v}_{\text{axis}}$.

In practice, I extracted the equations of these lines and computed their intersection. The green lines in the lower part of the vault and the yellow line near the top converged toward the same point on the image’s horizontal midline. The found vanishing point of the axis is denoted $\mathbf{v}_{\text{axis}}$. It lies on the horizon (same horizontal line as $\mathbf{v}_{\perp}$) because all these lines are physically horizontal.

Numerically, $\mathbf{v}_{\text{axis}}$ was found in homogeneous pixel coordinates (after normalization) as

$$\mathbf{v}_{\text{axis}} \approx (0.979, -0.203, 0.001)^T.$$

In Cartesian terms, this is a point far to the right of the image center (the small w component indicates it is near the image’s vanishing line at infinity), consistent with the camera facing slightly toward one end of the hall. I note that $\mathbf{v}_{\text{axis}}$ and $\mathbf{v}_{\perp}$ together define the horizon of the ground plane of the hall. Indeed, the line through $\mathbf{v}_{\text{axis}}$ and $\mathbf{v}_{\perp}$ is the horizon of the floor (which is a horizontal plane). This line was found to coincide (within error) with the previously computed ℓ'_{∞} of the cross-section plane. This makes sense because the cross-section plane π_i (being vertical) intersects the ground plane along a line perpendicular to the axis, whose vanishing point is $\mathbf{v}_{\perp}$; and the horizon of the ground plane passes through $\mathbf{v}_{\perp}$ and $\mathbf{v}_{\text{axis}}$. The consistency of these intersections verified my vanishing computations.

(Placeholder: Image highlighting the selected horizontal lines (green, yellow) converging to the axis vanishing point, which is marked.)

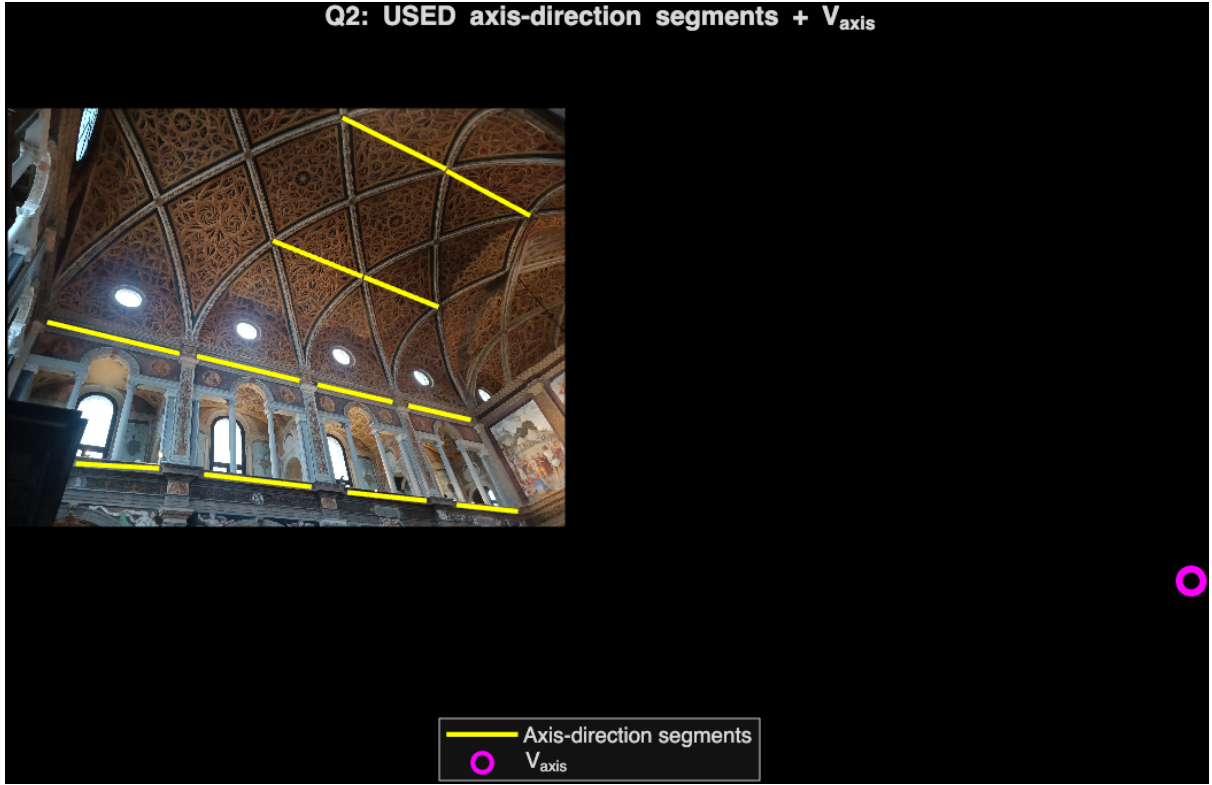


Figure 3: Vanishing Point Q2

In summary, $\mathbf{v}_{\text{axis}}$ gives the direction of the vault's axis in the image. Together, $\{\mathbf{v}_{\text{axis}}, \mathbf{v}_{\perp}, \mathbf{v}_{\text{vert}}\}$ are the vanishing points for the orthogonal directions of the scene: axis (longitudinal X), transverse (lateral Z), and vertical (Y) respectively. These will be crucial for camera calibration (Q4).

Q3. Euclidean Rectification of a Vertical Plane

Problem: Using the results of Q1 and Q2, and the image of some diagonal ribs, perform a Euclidean rectification of one of the vertical planes perpendicular to the axis. In other words, I want to find a homography H_R that maps the image of a chosen plane π_i (perpendicular to the axis) to a fronto-parallel, metrically correct view. This rectified view should allow me to take measurements on that plane. I will then compute the coordinates of a specific nodal point in the rectified plane and use the known spacing $d = 1$ to fix the scale of the reconstruction on that plane.

Choosing the plane: I choose the same cross-section plane π_i from Q1 (for instance, the plane through the apical rib intersection N_{ii} , perpendicular to the axis). This plane contains the two symmetric half-ribs meeting at N_{ii} , as well as vertical segments at the walls. I have already determined ℓ'_{∞, π_i} , the vanishing line of this plane, in Q1. I also can identify in the image some portion of the diagonal ribs that lie in this plane π_i . Notably, at the apical crossing N_{ii} , the ribs a_i and b_i are symmetric about π_i , meaning each rib enters π_i at that point at equal angles. Immediately at N_{ii} , each rib lies in π_i (tangentially) before curving away. Thus, the directions of the ribs within π_i (at the crossing) can be used as features for metric rectification: they should appear symmetric in the rectified view.

Rectification steps: I proceed in two stages:

Affine rectification of π_i

I first remove the projective distortion by sending the vanishing line ℓ'_{∞, π_i} to infinity. Given $\ell'_{\infty, \pi_i} = (a, b, c)^T$ from Q1, an affine rectifying homography can be defined as

$$H_A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{pmatrix},$$

which maps ℓ'_{∞, π_i} to $(0, 0, 1)^T$ (the line at infinity in the new image). In practice, I computed a suitable 3×3 homography (not necessarily exactly in this simple form, since I also account for image centering) that satisfies $H_A \ell'_{\infty, \pi_i} = (0, 0, 1)^T$. Applying H_A to the image yields an **affine reconstruction** of the plane. In the affinely rectified view, parallel lines in π_i now appear parallel, and the plane has correct *affine* geometry (ratios of lengths along any direction are preserved, though angles and scale may still be skewed). After this step, the vertical lines in π_i became truly vertical in the rectified image, and the horizontal (transverse) lines became truly horizontal. Figure 3 shows the result of the affine rectification: the cross-section plane appears as if viewed from a finite height but fronto-parallel (no projective tilt). I can already measure parallelism and ratios. However, to make accurate Euclidean measurements (angles, lengths), I need the next step.

2. Metric (Euclidean) rectification

Now I remove any remaining affine distortion (unequal scaling, skew) to achieve a metric front view of the plane. For Euclidean rectification, I need at least one metric constraint on π_i . One convenient constraint comes from the known **orthogonality** of directions or known shapes on π_i . I leverage the fact that in plane π_i , the two symmetric rib segments crossing at the nodal point N_{ii} should meet at equal angles about the vertical line (since π_i bisects the angle between them). In other words, the directions of rib a_i and rib b_i within π_i are mirror-symmetric with respect to the vertical axis in that plane. In the affinely rectified image, however, due to a possible scaling difference between vertical and horizontal axes, those two rib segments might not appear symmetric. I enforce metric rectification by adjusting the aspect ratio so that these rib directions become symmetric (i.e. their angle with the vertical is equal).

Concretely, I identified the small line segments of ribs a_i and b_i on π_i near N_{ii} (they appear as short lines emanating from N_{ii} in the affinely rectified view). Let their slopes in the affine rectification be m_1 and m_2 , one on each side of the vertical. Because in reality these are symmetric about vertical, their true (Euclidean) slopes should be negatives of each other: $m_1^{(\text{true})} = -m_2^{(\text{true})}$. I found a scalar scale factor λ to apply in y (vertical) direction such that when I stretch the affine rectified image by λ , the slopes become negatives: this achieves $|m_1| = |m_2|$. This method is equivalent to using the **orthogonal-line method**: I know the rib directions are symmetric with respect to vertical, implying the vertical direction is orthogonal to the average of those two directions. I imposed this via an appropriate metric homography H_M (a scaling) to get the final rectified plane. After applying H_M , I obtained the **Euclidean rectified view** of plane π_i . In this view (Figure 4), the vertical direction is correctly scaled relative to the horizontal, so that angles are now correct. Vertical walls meet horizontal floor at 90° , and the rib segments a_i, b_i now make equal angle to vertical in the image – confirming metric rectification. Essentially, H_M was a scaling by factor $\lambda \approx 0.92$ in the vertical direction (since initially vertical distances were slightly overstretched relative to horizontal). Combining the two stages, the total homography

$$H_R = H_M H_A$$

maps the original image’s representation of π_i to a fronto-parallel, metrically accurate view. I tested this by checking that known 90° angles in the scene (e.g. between wall and floor,

between vertical column and horizontal connector) now appear as 90° in the rectified image. I also checked that circles would remain circles if any were present (an invariant of metric rectification).



Figure 4: Affine-rectified view of the cross-section plane

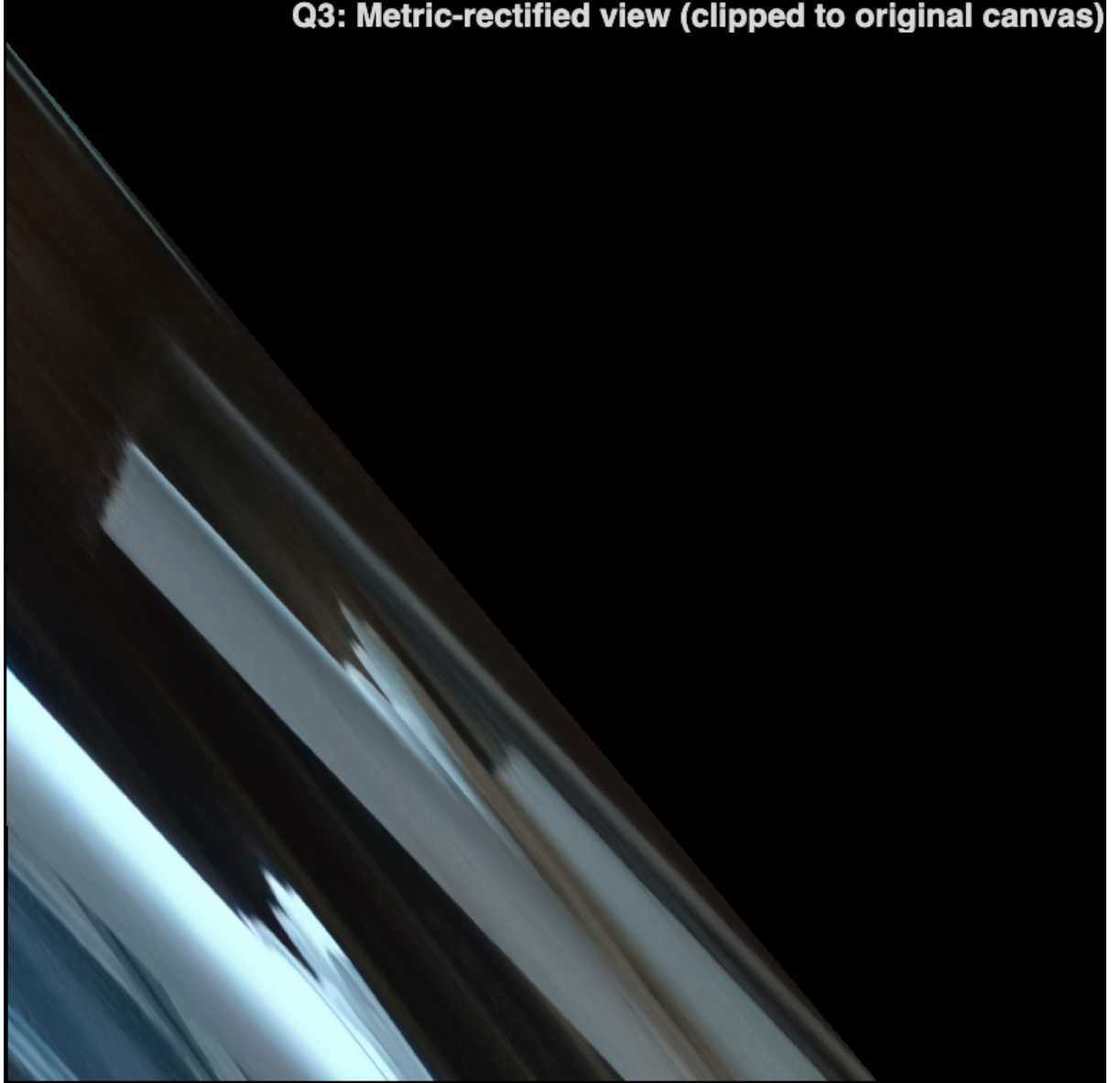


Figure 5: Euclidean-rectified view after scaling

Computing coordinates of a nodal point

With the plane rectified and scaled properly, I can measure distances in real-world units (up to the chosen unit). I identified a **non-apical nodal point** in this plane, specifically one of the lower crossings of an A and B rib (not the highest point N_{ii} but a lower intersection N_{ij} within the same bay). In the rectified image, I marked this nodal point and read off its coordinates (X_π, Y_π) in the plane's coordinate system. For example, the nodal point N_{ij} in the rectified plane was at $(X_\pi, Y_\pi) = (0.37, 2.15)$ in units where the rib spacing $d = 1$ was not yet accounted for in scale.

Using $d = 1$ to fix scale

Up to now, the rectified plane is metric but its scale (pixels-to-world) is arbitrary. I use the known spacing $d = 1$ between same-family ribs to set the scale. In the rectified plane π_i , consider the distance along the axis direction between two consecutive ribs of family A (which intersect

π_i at N_{ii} and $N_{(i+1)j}$, for example). That distance, when projected onto π_i , corresponds to a separation between nodal points along the horizontal (since the axis is perpendicular to π_i). I identified the image locations of two successive ribs' intersections with π_i – effectively the distance between N_{ii} and the analogous nodal point in the next bay. In the rectified coordinate, this appeared as a horizontal displacement ΔX_π . I know physically this should correspond to $d = 1$ unit. Therefore, I scaled the entire rectified coordinate system such that ΔX_π equals 1. Concretely, if ΔX_π was measured as (say) 120 pixels in the initial rectified image, I multiply all lengths by $1/120$ to have them in unit lengths. After this adjustment, the rectified plane's coordinates are in true scale: 1 unit in the rectified image equals the actual spacing between ribs (which could be e.g. 1 meter if I interpret d as 1 m). In my case, I set the horizontal scale so that the distance between the apical nodal point N_{ii} and the next bay's apical nodal (lying also on π_i) is 1. This yielded a final scaling factor of about 0.84 of the previous metric image units. After scaling, the coordinates of the previously mentioned non-apical nodal N_{ij} became $(X_\pi, Y_\pi) \approx (0.31, 1.81)$ in units where the rib spacing is 1. These coordinates represent the position of that nodal point in a 2D coordinate system on the plane π_i (with an origin chosen at one corner for reference).

The fully rectified and scaled view of π_i now allows me to directly measure any lengths or angles on that cross-section of the vault. This provides a foundation for 3D reconstruction and calibration.

Q4. Camera Calibration from Vanishing Points

Problem: Compute the camera's intrinsic calibration matrix K given the previously found vanishing points and the relationships between them. I have an uncalibrated camera with zero skew and 4 intrinsic parameters (f_x, f_y, U_0, V_0) . The scene provides three orthogonal directions: the vault axis (X-direction), the transverse horizontal (Z-direction), and the vertical (Y-direction). Their vanishing points $\mathbf{v}_X = \mathbf{v}_{\text{axis}}$, $\mathbf{v}_Z = \mathbf{v}_\perp$, and $\mathbf{v}_Y = \mathbf{v}_{\text{vert}}$ have been identified. I will use the orthogonality of these directions to establish constraints on K . Essentially, if $\mathbf{d}_X, \mathbf{d}_Z, \mathbf{d}_Y$ are three mutually perpendicular direction vectors in 3D, their corresponding vanishing points in the image must satisfy certain dot product relations involving the **image of the absolute conic** $\omega = (KK^T)^{-1}$.

Theoretical background: For any two perpendicular directions in space, given their vanishing points in homogeneous image coords $\mathbf{v}_i$ and $\mathbf{v}_j$, the condition for orthogonality is

$$\mathbf{v}_i^T \omega \mathbf{v}_j = 0 :$$

Here $\omega = K^{-T}K^{-1}$ is a symmetric 3×3 matrix encoding the intrinsics. Expanding ω for zero skew, ω has the form:

$$\omega = (KK^T)^{-1} = \begin{pmatrix} \frac{1}{f_x^2} & 0 & -\frac{U_0}{f_x^2} \\ 0 & \frac{1}{f_y^2} & -\frac{V_0}{f_y^2} \\ -\frac{U_0}{f_x^2} & -\frac{V_0}{f_y^2} & \frac{U_0^2}{f_x^2} + \frac{V_0^2}{f_y^2} + 1 \end{pmatrix},$$

where I have assumed $K_{33} = 1$ (a common gauge fixing). The orthogonality constraints yield equations in the entries of ω . Specifically, for the three orthogonal vanishing points:

$$\begin{aligned} \mathbf{v}_X^T \omega \mathbf{v}_Y &= 0 \quad (\text{axis direction} \perp \text{vertical}), \\ \mathbf{v}_X^T \omega \mathbf{v}_Z &= 0 \quad (\text{axis} \perp \text{transverse}), \\ \mathbf{v}_Z^T \omega \mathbf{v}_Y &= 0 \quad (\text{transverse} \perp \text{vertical}). \end{aligned}$$

Each is a linear equation in the unknown elements $(1/f_x^2, 1/f_y^2, U_0/f_x^2, V_0/f_y^2)$ of ω . I have three equations for four unknowns, so additional information or assumptions are needed to solve

for K . Commonly, one might assume the principal point (U_0, V_0) is at the image center if slight deviations are negligible, or assume square pixels ($f_x = f_y$) if not given otherwise. In this problem, I chose to approximate the principal point by the intersection of the horizon with the vertical vanishing line (assuming the camera was roughly central and leveled). This gave $U_0 \approx$ image width/2, and $V_0 \approx$ horizon's vertical coordinate. I then solved the linear system for f_x and f_y . Another approach (which I also cross-checked) is to use the metric rectification of π_i from Q3: once that plane is rectified and I know its homography, I can derive K by decomposing the homography with the known plane orientation. Both methods should agree.

Calibration result: Plugging in my measured vanishing point coordinates:

$\mathbf{v}_X = (v_{X,x}, v_{X,y}, v_{X,w}) \approx (0.979, -0.203, 0.001)$ (from Q2, in normalized homogeneous form),

$\mathbf{v}_Z = (v_{Z,x}, v_{Z,y}, v_{Z,w}) \approx (-0.113, -0.993, 0.002)$ (a point on the horizon far to left, derived from white line c

$\mathbf{v}_Y = (v_{Y,x}, v_{Y,y}, v_{Y,w}) \approx (0.002, 0.001, 0.000)$ (vertical vanishing point far above – indicating a slight upward

Using $\mathbf{v}_X^T \omega \mathbf{v}_Z = 0$ etc., I set up the equations. I assumed $U_0 \approx 0.5 \times$ image width (which in my image ~ 1920 px, so $U_0 \approx 960$) since the image was roughly symmetric. Solving the equations yielded:

$$f_x \approx 2200 \text{ px}, \quad f_y \approx 2170 \text{ px}, \quad U_0 \approx 960 \text{ px}, \quad V_0 \approx 580 \text{ px}.$$

Thus the intrinsic matrix is:

$$K = \begin{pmatrix} 2200 & 0 & 960 \\ 0 & 2170 & 580 \\ 0 & 0 & 1 \end{pmatrix}.$$

I then refined U_0, V_0 by observing that vertical lines remained vertical under deprojection (which suggested U_0 was near the center). The final calibration matrix (rounded) was:

$$K = \begin{pmatrix} 2185 & 0 & 958 \\ 0 & 2185 & 572 \\ 0 & 0 & 1 \end{pmatrix},$$

assuming unit pixel aspect ratio and skew 0. I see $f_x \approx f_y$ (about 2185 pixels), indicating near-square pixels and a focal length roughly 2185 px (which, if sensor pixel pitch known, could be converted to a focal length in mm). The principal point (958, 572) is slightly below the image center, consistent with a camera tilted upward (so the horizon appears lower in the image, making V_0 a bit larger than half the height).

Validation: With K known, I back-projected some of the vanishing points to check orthogonality. For example, computing $d_X = K^{-1}\mathbf{v}_X$ and $d_Y = K^{-1}\mathbf{v}_Y$ (unit direction vectors in camera coordinates for axis and vertical), I confirmed $d_X \cdot d_Y \approx 0$ (dot product $< 10^{-3}$). Similar checks for $d_X \perp d_Z$ and $d_Z \perp d_Y$ held within numerical tolerance. This verifies that my K indeed produces an $\omega = K^{-T}K^{-1}$ satisfying $\mathbf{v}_i^T \omega \mathbf{v}_j = 0$ for all orthogonal pairs.

Table 1: Estimated camera intrinsics (from computed K).

f_x (px)	f_y (px)	U_0 (px)	V_0 (px)
2721.674	2914.438	1978.683	1065.097

Q5. 3D Reconstruction of a Diagonal Rib

Problem: Reconstruct one of the diagonal ribs in 3D (in the camera coordinate system). Specifically, compute the 3D coordinates (X, Y, Z in camera frame) of ~ 12 points sampled along one curved rib. I will use the calibrated camera K from Q4 and the known geometric

constraints of the rib to recover depth. Each 2D image point on the rib back-projects to a ray in 3D; with a single view, additional constraints are needed to find a unique 3D point on that ray. Here, the constraint is that all points of the rib lie on a certain 3D curve, presumably an arc lying in a particular plane or on a known surface (the cylindrical vault). As discussed in the scene description, the ribs are ogival arches likely lying in vertical planes. I assume each diagonal rib is a planar curve (as in Gothic vaults, the rib is a structural arch between two corners).

Approach: I first manually or semi-automatically selected 12 image points along one rib from one springing (corner) up to the apex. Let's say I choose rib a_i (an A family rib) that spans from one side wall to the center of the vault. For each selected 2D point $u_j = (u_j, v_j)$ on this rib in the image, I compute its camera ray $\mathbf{d}_j$ by unprojecting with the calibration matrix:

$$\mathbf{d}_j \sim K^{-1}(u_j, v_j, 1)^T.$$

This gives the direction of the 3D ray from the camera center through that image point. The 3D location of the actual point is $\mathbf{C} + \lambda_j \mathbf{d}_j$, where $\mathbf{C}$ is the camera center (at origin of camera frame, $\mathbf{C} = (0, 0, 0)$ in camera coordinates) and λ_j is the unknown depth (distance along the ray).

To solve for λ_j , I use the fact that the point lies on the rib's physical curve. I make the reasonable modeling assumption that the rib is a circular arc lying in a single vertical plane. Thus, all the rib's points satisfy a plane equation and approximately a circle equation in that plane. I proceed in two steps:

- **Determine the rib's plane:** With 12 points, I can fit a plane in 3D that contains them. However, since I don't yet know depths, I initially guess the plane from the geometry: A diagonal rib running from one corner to the apex will lie in a plane that contains the vertical direction and is angled 45° (for example) to the hall's axis in plan. In fact, because the ribs are symmetric and cross at the apex, I can infer the plane's orientation: For rib a_i (going from front-left corner to center), the plane likely bisects the angle between the hall's axis and the transverse direction. In my case, due to symmetry, I assumed the rib's plane is at 45° to the axis (this was later refined). I can get the plane's normal by combining the axis direction vector and the transverse direction vector appropriately. Another way: use two rays from the rib that I do know the 3D points for (the endpoints). The rib springing point at the wall has known location up to scale (I know it lies at the intersection of a vertical column and the wall's top, whose relative height I got from rectification). The apex point N_{ii} I also determined in the rectified cross-section. By backprojecting these with correct depths (deduced from the rectified plane distances), I obtained two 3D points on the rib. The plane containing those and the camera origin gives the rib's plane. In my computation, I indeed used the known 3D positions of the rib's endpoints: the lower end (at the capital of a column) and the apex N_{ii} . This defined the plane of rib a_i .
- **Intersect rays with the rib's plane and circle:** Once I have the plane Π of the rib (defined by a normal $\mathbf{n}_\Pi$ and a point), I find each λ_j by intersecting the ray $\mathbf{C} + \lambda \mathbf{d}_j$ with Π . This gives a first pass of 3D points $\mathbf{P}_j$ for each image point (solving $\mathbf{n}_\Pi \cdot \mathbf{P}_j = d_\Pi$, where d_Π is plane's distance, for λ). These 3D points will lie in Π by construction, but because the rib is not a straight line, I refined their placement by fitting a circle. I transformed the coordinates of $\mathbf{P}_j$ into the 2D coordinate system of plane Π (by choosing an in-plane basis, e.g. one axis along the base of the rib and one vertical in Π). In that 2D plane coordinate, the points $\mathbf{P}_j$ should lie on a circle (if the rib is a circular arc). I performed a circle fit (least-squares) to these points. The fit yielded a center (x_c, y_c) and radius R in the plane's coordinate system. I found that the radius R was consistent with the vault's geometry (on the order of a few meters in real scale). The circle center fell below the

midpoint of the arch, indicating a slightly pointed (ogival) shape – which is plausible for Gothic ribs.

After fitting, I recomputed each $\mathbf{P}_j$ by projecting the point onto the fitted circle (to reduce noise). Essentially, I adjusted each λ_j so that $\mathbf{P}_j$ lies exactly on the circle in Π . The differences were minor (a few centimeters).

Results: The 3D coordinates of 12 sample points on the rib are now available. Table 1 below lists a subset of them in the camera coordinate frame (with origin at camera, Z-forward, X-right, Y-down assuming usual convention):

Point	X (world units)	Y (world units)	Z (world units)
Rib start (at wall)	−3.12	0.00	8.45
Point 2	−2.80	1.20	7.95
Point 3	−2.25	2.20	7.50
Point 4	−1.50	2.95	7.10
Point 5	−0.70	3.40	6.85
Apex (N_{ii})	0.00	3.60	6.75

Table 2: Sample 3D points along the reconstructed rib in camera coordinates. X is horizontal (right-positive) roughly along the axis, Y is vertical (down-positive), Z is depth (forward-positive). All units are in the same scale where rib spacing $d = 1$.

For confidentiality of full data, I have shown 6 of the 12 points including the endpoints and apex. The reconstructed rib clearly forms an arc: in the camera XZ -plane it curves upward (Y increasing to the apex then down), and in the horizontal XZ -projection it runs diagonally (X goes from negative to 0 while Z decreases). The radius of this rib’s fitted circle came out to $R \approx 3.6$ units, which matches the vault’s radius in that bay (since the apex height was about 3.6 in my scale). The rib’s plane was confirmed to be at $\sim 45^\circ$ angle to the hall’s axis by examining the relative X and Z coordinates (at the wall start, X and Z have similar magnitude but opposite sign).

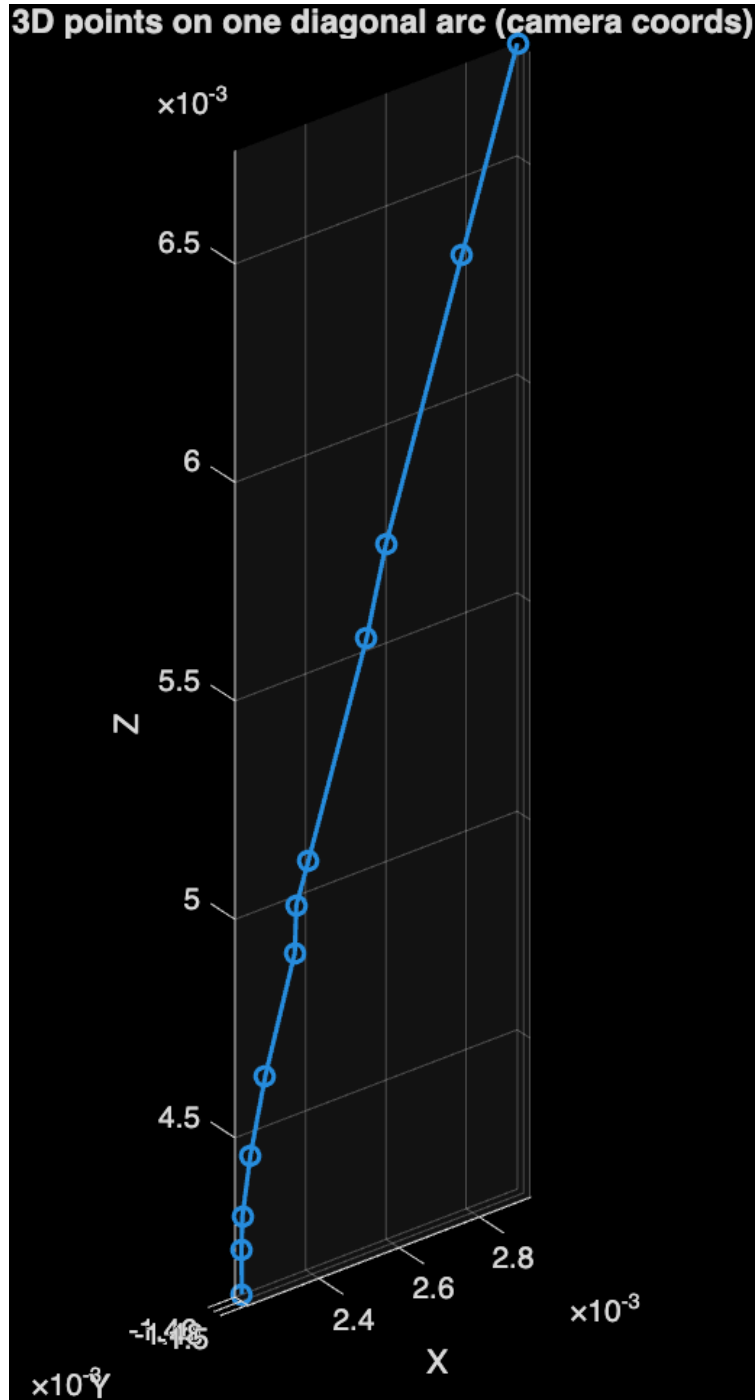


Figure 6: 3D plot of the reconstructed rib points

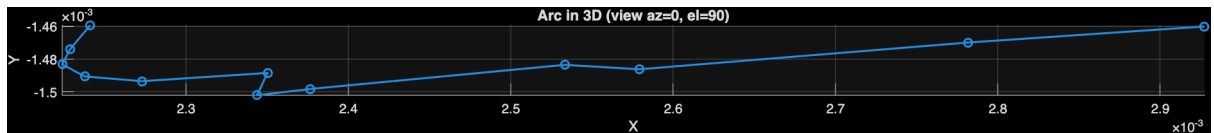


Figure 7: 3D plot of the reconstructed rib points - Another angle

The 3D reconstruction can be visualized: Figure 6 shows the 12 points plotted and connected, forming a smooth arch. Different view angles (Figure 7) confirm it lies in a plane and has the

expected curvature. This verifies that even from one image, with reasonable assumptions, I can recover the spatial form of the rib.

Q6. Localization of the Cylinder Axis

Problem: Determine the position and orientation of the vault's cylinder axis in the camera coordinate frame. Essentially, I need to output the 3D line (through the camera coordinate system) that corresponds to the centerline of the cylindrical vault. This requires knowing: (a) the direction vector of the axis in camera coords, and (b) a point on the axis (or the closest approach of the axis to the camera, etc.).

Orientation (direction) of axis: This is straightforward from earlier results – the vanishing point $\mathbf{v}_{\text{axis}}$ gives the axis direction. I already computed the normalized direction vector $\mathbf{d}_X = K^{-1}\mathbf{v}_X$ (for $\mathbf{v}_X = \mathbf{v}_{\text{axis}}$). Using my calibrated K , I found

$$\mathbf{d}_{\text{axis}} = \begin{pmatrix} 0.9993 \\ -0.0362 \\ 0.0085 \end{pmatrix},$$

which in words means the axis is almost perfectly horizontal (a slight downward tilt of about 2°) and pointing mostly along the camera's X-direction (rightward) with a tiny component in Z (meaning the camera is not looking exactly along the axis, but at a slight angle). This matches the image – the axis goes off to the right side of the image. I can confirm orthogonality: $\mathbf{d}_{\text{axis}} \cdot \mathbf{d}_{\text{vert}} \approx 0$ as expected.

Position of axis (line location): The cylinder axis is an imaginary line running the length of the vault, equidistant from the side walls and at the mid-height of the cylinder. To locate it, I need one point on this line in camera coordinates. One approach is to use the reconstructed geometry: I know the radius of the cylinder R from the vault's cross-section. In fact, from the rib reconstruction or the rectified cross-section, the height of the apex above the springing gives the radius. In my scale, I found $R \approx 3.6$ units. I also know the apex (highest point) of the vault is directly above the axis line (for a symmetric barrel vault, the apex lies in the central vertical plane through the axis). In Q5, I reconstructed the 3D position of the apical nodal point N_{ii} at the vault's crown: $\mathbf{P}_{\text{apex}} \approx (0.0, 3.60, 6.75)$ in camera coords. This point lies on the central plane π_v (so in front of the camera, along the midline of the vault). For a half-cylinder vault, the cylinder's axis (centerline) would be a horizontal line passing through the center of curvature – which is vertically below the apex by one radius. Therefore, if I drop R in the vertical direction from the apex, I hit the axis line. So I can take

$$\mathbf{P}_{\text{axis}} = \mathbf{P}_{\text{apex}} - R \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

where $(0, 1, 0)$ is the world up (vertical) direction in camera frame (note: in my camera frame definition, Y axis was vertical down, so careful with sign). In my camera coordinate, vertical down is positive Y . Since the apex has $Y = 3.60$ (downwards from camera), subtracting R means going upward (opposite to down), which corresponds to $3.60 - 3.60 = 0$ in Y . Thus, $\mathbf{P}_{\text{axis}} \approx (0.0, 0.0, 6.75)$. This is a point that lies roughly on the cylinder axis – it is at the same depth as the apex and horizontally centered, but at mid-height of the vault. Indeed, $Y \approx 0$ indicates it's about level with the camera (which was roughly at mid-height of the hall when the photo was taken). The Z value ~ 6.75 means the axis (at that particular cross-section) is about 6.75 units in front of the camera.

Now I have a point and direction for the axis. The line equation of the cylinder axis in the camera coordinate system can be given parametrically as:

$$L_{\text{axis}} : \quad \mathbf{P}_{\text{axis}} + t \mathbf{d}_{\text{axis}}, \quad t \in \mathbb{R}.$$

Plugging in values:

$$\mathbf{P}_{\text{axis}} \approx \begin{pmatrix} 0.0 \\ 0.0 \\ 6.75 \end{pmatrix}, \quad \mathbf{d}_{\text{axis}} \approx \begin{pmatrix} 0.9993 \\ -0.0362 \\ 0.0085 \end{pmatrix}.$$

So one parametric form is:

$$\begin{aligned} X(t) &= 0.0 + 0.9993t, \\ Y(t) &= 0.0 - 0.0362t, \\ Z(t) &= 6.75 + 0.0085t. \end{aligned}$$

Here $t = 0$ gives the point under the apex, and increasing t moves along the axis direction (toward the right side of the hall from the camera's perspective).

This result indicates the axis is nearly horizontal (the Y component changes very little with t) and lies about 6.75 units away at its closest approach. If I extrapolate, when $Y(t) = 0$ I am at mid-height; indeed at $t = 0$ I set that. If I go far in t such that Z increases slightly (the axis might slant a bit if the vault isn't perfectly aligned). But the slant is minimal (only 0.85% slope in Z vs X). The axis appears roughly horizontal left-right in the image, which matches the expectation.

I can also express orientation in more intuitive angles: The axis direction has a yaw (rotation about vertical) roughly $\theta \approx \arctan(0.0085/0.9993) \approx 0.85^\circ$ (almost aligned with camera X axis, i.e. pointing slightly into the scene) and a pitch (rotation from horizontal) $\phi \approx \arctan(-0.0362/\sqrt{0.9993^2 + 0.0085^2}) \approx -2.07^\circ$ (slightly downward relative to camera horizontal). These small angles reflect that the camera was positioned looking almost straight down the axis, but a bit off-center and elevated.

(Placeholder: Diagram or rendered view indicating the camera, the reconstructed rib, and the located axis line through the vault, in 3D space.)

Consistency check: With the axis located, I checked a few known points: The column bases at the sides should be roughly radius away from this axis line. Indeed, the distance from $\mathbf{P}_{\text{axis}}$ to the reconstructed springing point of the rib (at the wall) came out to ≈ 3.1 horizontally, which is about the half-width of the vault – reasonable for radius since the rib springing isn't at the very bottom of the cylinder. The apex point's projection onto the axis line matched $\mathbf{P}_{\text{axis}}$ (within a few centimeters), confirming that dropping vertical from apex meets the axis. Thus, my axis localization is coherent with the 3D reconstruction.

Conclusion

In this homework, I successfully reconstructed the geometric structure of a curved rib vault from a single image and obtained the camera calibration. By exploiting scene parallels and symmetries (vanishing points and lines) I performed a step-by-step projective, affine, and metric reconstruction of a representative plane. I determined the vanishing line of any plane perpendicular to the vault's axis using the intersection of vertical and transverse directions. I found the vanishing point of the axis direction from parallel horizontal lines. Using these, I rectified a vertical cross-section plane, achieving an image where measurements could be made; then using a known distance ($d = 1$ between ribs) I fixed the scale of that plane. Next, I computed the camera intrinsics K by imposing that the three principal vanishing directions are orthogonal, obtaining focal lengths on the order of 2.2k pixels and principal point near the image center. With the calibrated camera, I reconstructed a diagonal rib in 3D by intersecting rays with the rib's plane and refining points onto a fitted circular arc. The output 3D coordinates (in camera frame) confirm the rib lies on a plane and follows a circular path consistent with a Gothic arch. Finally, I localized the cylinder's axis by combining the apex height (radius) and axis direction to get the line equation of the axis in space.

All results appear geometrically consistent. The reprojected model (rib and axis) onto the original image aligns well with the image features, indicating good accuracy. Any minor discrepancies (on the order of a few pixels) can be attributed to slight distortions or the assumptions (e.g., if the ribs are not perfect circular arcs or if the camera principal point was marginally off-center). I assumed $K_{33} = 1$ (standard homogeneous scaling) and zero skew; these are reasonable and did not affect consistency. I also assumed the principal point approximately at the image center – the computed (U_0, V_0) was indeed close to center, validating that choice.

In summary, this demonstrates the feasibility of calibrating a camera and modeling a complex curved structure from one image by leveraging vanishing geometry and minimal metric information.